# Same Deadline, Different Persistence

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Data Mining Course Project

## Question

Within the same module presentation, among students whose clicks after the first TMA deadline are comparable, are submission timing and post-deadline persistence associated with failing rather than passing?

## Design

Anchor: the deadline of the first positive-weight TMA. Windows: 14 days before and 14 days after. Fail is the positive class; Withdrawn is described separately. Pairs: 1:1 within presentation on post-deadline clicks, 5% tolerance, zero clicks only with zero clicks.

## Evidence

Cohort: 24,125 eligible records, 19,806 in the binary comparison, 6,092 Fail. Paired differences (Fail − Pass): submitted_by_cutoff: -0.124; submitted_after_due: +0.084; relative_submission_day: +1.381; post_active_days: -0.368; post_longest_inactive_run: +0.389; post_week2_share: -0.037. Model M2 Fail AP 0.661 against a no-skill floor of 0.308.

## Reading

The pairing conditions on post-deadline volume, so it cannot estimate the overall failure rate. Persistence separates outcomes descriptively far more clearly than it improves prediction, and the score model is a retrospective upper bound.

LaTeX TikZposter